

# Selecting An ELECTRIC MOTOR And DRIVE SYSTEM



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**E**lectric motors and drives are important components of industrial automation. Their selection therefore is an important step in the process of building an automated system. You as a buyer must know what types of motors and drives are available and suitable for your application.

The most common electric motors are meant for machine tools, special motors for industrial robots, blowers, fans, power tools and pumps. In industry, motors and drives powering mechanical equipment account for about 65 to 70 per cent of the total energy consumed. Reducing this figure is of prime importance for developing energy-saving systems. The longer the drives and motors are in operation, the higher the savings required.

Broadly, there are two types of motors: AC motors and DC motors. Within these categories, there are sub-divisions and various variations.

## AC motors

Here, the current is alternating, and so the motor will run but only at sine-wave frequency. Given below are the two types of AC motors:

**Asynchronous induction motors.** Induction motors are regularly used for constant-

speed and adjustable-speed drive applications. These are popular for automated industrial applications including general-purpose and inverter/vector duty. Two variations of induction motors are single- and three-phase.

**Synchronous AC motors.** These are three-phase AC motors that run at synchronous speed without a slip. Unlike induction motors, these are not self-starting, and have to be run up to synchronous speed by some means before these can be synchronised to the supply.

These motors are used when load keeps changing and speed is critical. These are mainly used in constant-speed applications. With inverters and cycle converters, these can be in used in variable-speed applications.

**Single-phase AC motors.** These are small-size motors used in domestic applications like fans, hair dryers, washing machines, vacuum cleaners, kitchen equipment, blowers, small power tools, dairy machinery, small farming equipment and so on.

There are some special types of induction motors, construction wise, built for applications in different working conditions.

**Wash-down motors,** for instance, are designed for applications where a lot of water is splashed or directed on and over motors. These are suitable for food processing, pharmaceuticals and packaging applications.

**Hazardous-duty motors** are used where the potential for igniting the environment exists. These are explosion-proof since their design contains provision for any internal failure within the body of the motor. Classification of hazardous areas falls under IS:5572:2009 and IS:5571:2009.

**Severe-duty motors** are designed for harsh industrial-processing applications. Features include protection against contamination, moisture, vibration and demanding duty-cycles. These are commonly found in

Fig. 1: AC torque motors

